

DARKTRACE: Results and Discussion

Explainable and Forensically Verifiable Multilingual Dark Web Threat Intelligence Framework

1 Results and Discussion

We report measured results for all seven components of DARKTRACE; every value in this section comes from executed experiments (seed = 42 unless noted). We first present threat-text classification (Sec. 1.1), then multilingual detection (Sec. 1.2), explainable severity scoring (Sec. 1.3), evidence integrity (Sec. 1.4), STIX export (Sec. 1.5), and the integration ablation (Sec. 1.6). Section 1.7 discusses whether the evidence is sufficient for an SCI/SCIE submission and what would strengthen it.

1.1 Threat-Text and Darknet Classification

On the CoDA dark-web text corpus, the TF-IDF+Linear-SVM configuration reaches a macro-F1 of 0.919 (accuracy 0.911, FPR 0.011), and TF-IDF+Logistic Regression 0.903 macro-F1 (Table 1). On the CIC-Darknet2020 traffic benchmark, HistGradientBoosting attains 0.794 macro-F1 with AUC 0.99, slightly ahead of Random Forest (0.777 macro-F1, AUC 0.962). The text task yields markedly higher macro-F1 than the traffic task, consistent with the richer lexical signal in forum/market text. Per-class F1 breakdowns are shown in Fig. 1 (generated for each model).

Table 1: Classification results (Phase 1). CoDA is dark-web text; CIC-Darknet2020 is encrypted-traffic. AUC is not defined (nan) for the multiclass one-vs-rest SVM/LogReg setup as configured.

Model	Dataset	Accuracy	Macro-F1	AUC	FPR
Random Forest	CIC-Darknet2020	0.8718	0.7766	0.9615	0.0182
HistGradientBoosting	CIC-Darknet2020	0.8746	0.7942	0.9900	0.0178
TF-IDF + LogReg	CoDA	0.8963	0.9034	—	0.0129
TF-IDF + LinearSVM	CoDA	0.9109	0.9192	—	0.0112

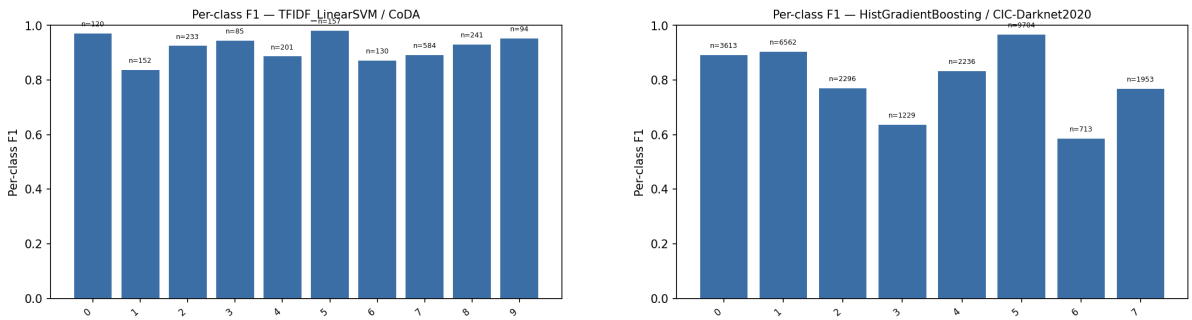


Figure 1: Per-class F1: CoDA text (left, Linear-SVM) and CIC-Darknet2020 traffic (right, HistGradientBoosting).

1.2 Multilingual Detection

Table 2 and Fig. 2 report multilingual performance under two regimes: *pooled* multilingual training and *zero-shot* transfer from English. Under pooled training the model is strong on the well-represented languages (EN macro-F1 0.908, $N=1757$; RU 0.916, $N=114$) and on small samples reaches high but less reliable values (PT 1.000, $N=10$). Under zero-shot transfer, English is essentially preserved (0.916) but transfer to RU (0.436), DE (0.443), and FR (0.516) shows the expected cross-lingual penalty. A separate cross-domain evaluation (Table 3) using a transformer trained on social-media data generalizes well to Hindi (0.958 macro-F1, $N=1146$) but only moderately to Arabic (0.697, $N=1778$), reflecting the known difficulty of Arabic morphology in low-resource settings.

Table 2: Multilingual detection (Phase 2): pooled vs. EN→ zero-shot transfer.

Language	N	Macro-F1 (Pooled)	Macro-F1 (Transfer)
EN	1757 / 1768	0.9078	0.9160
RU	114 / 542	0.9159	0.4357
FR	25 / 99	0.7439	0.5157
DE	24 / 129	0.6528	0.4425
ES	15 / 61	0.6787	0.5727
PT	10 / 54	1.0000	0.6828

Table 3: Cross-domain low-resource evaluation (Phase 2b), transformer model.

Language	N_{test}	Macro-F1	Accuracy	Domain
Hindi (hi)	1146	0.9580	0.9581	cross-domain (social media)
Arabic (ar)	1778	0.6967	0.7340	cross-domain (social media)

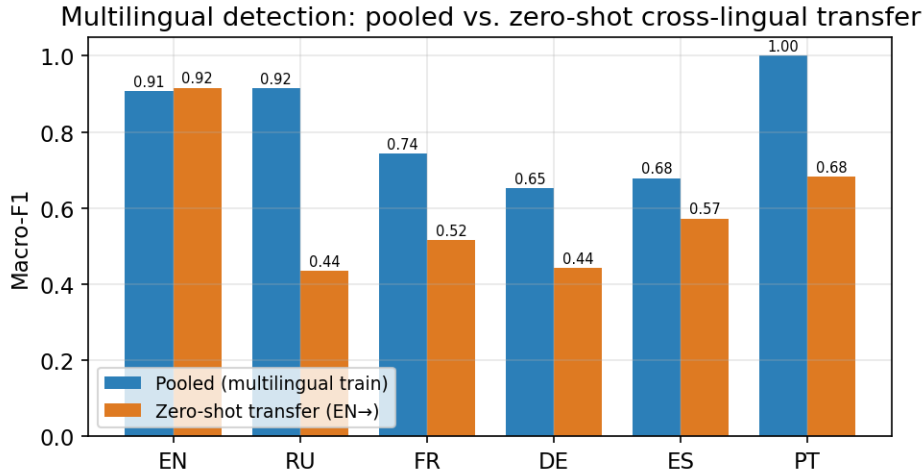


Figure 2: Pooled vs. zero-shot cross-lingual transfer macro-F1. The transfer gap is small for EN and large for RU/DE/FR, motivating few-shot adaptation for low-resource languages.

1.3 Explainable Severity Scoring

The explainable ensemble was trained on 7986 items and tested on 1997, scoring a 4-level severity mapping (0–3) derived from threat category. It achieves AUC 0.9752, average precision 0.9757, NDCG@10 0.9969, and Brier 0.0537, while additionally providing SHAP and LIME attributions

with a measured faithfulness gap of 0.0506 and stability (Jaccard) of 0.299 (Table 4, Figs. 3). The black-box MLP baseline is marginally higher on ranking (NDCG@10 1.000) but worse calibrated (Brier 0.0754) and provides *no* faithful explanation. DARKTRACE thus buys interpretability and better calibration at a negligible ranking cost—a favorable trade-off for investigator-facing use. The relatively low explanation stability (0.299) is a limitation we flag for future work.

Table 4: Explainable severity scoring (Phase 3) vs. black-box baseline.

Model	AUC	AP	NDCG@10	Brier	Faith. gap	Stab. (Jaccard)
Explainable Ensemble	0.9752	0.9757	0.9969	0.0537	0.0506	0.299
Black-box MLP (baseline)	0.9707	0.9715	1.0000	0.0754	—	—

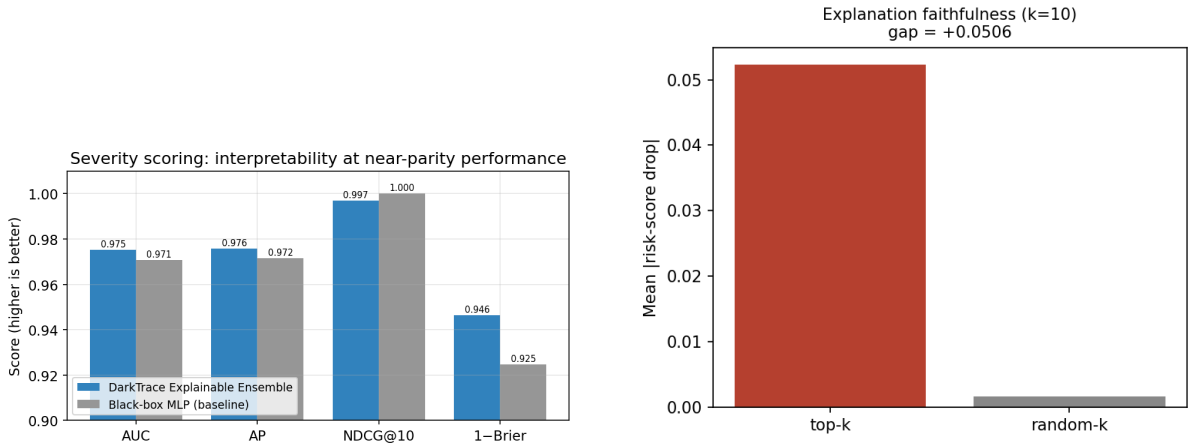


Figure 3: Left: interpretability–performance trade-off (explainable ensemble vs. black-box). Right: faithfulness of the explanations (perturbation-based).

1.4 Evidence Integrity

The forensic integrity layer was exercised over 500 evidence items (Table 5). Using a local hash-chain backend it sustains 8747.6 items/s with median (p50) sealing latency 0.094 ms and p95 0.233 ms. The full chain verified successfully (**Chain_verified=True**) and the adversarial tamper test was detected (**Tamper_detected=True**), demonstrating the intended tamper-evidence at negligible overhead. This supports the claim that forensic soundness can be a cross-cutting property of the pipeline rather than a costly add-on.

Table 5: Evidence-integrity results (Phase 4), local hash-chain backend.

Backend	<i>N</i>	Thr. (it/s)	p50 (ms)	p95 (ms)	Chain ok	Tamper det.
local_hash_chain	500	8747.6	0.094	0.233	True	True

1.5 Standards Export (STIX/TAXII)

The extraction layer exported 300 findings into a valid STIX 2.1 bundle of 301 objects including 300 indicators (**Bundle_valid=True**), built in 0.302 s (Table 6), confirming investigator-ready, standards-compliant output.

Table 6: STIX/TAXII export (Phase 5).

Export	Findings	STIX objs	Indicators	Bundle valid	Build (s)
STIX2.1/TAXII	300	301	300	True	0.302

1.6 Integration Ablation

The ablation measures *capability properties* of the emitted threat-intelligence object (label, severity, faithful explanation, sealed evidence, verifiable provenance, standards export) and a composite *actionability* score, over 300 findings. Only the full pipeline attains all properties and the maximum actionability of 0.955; removing any component reduces it—explanation $\rightarrow$ 0.833, scoring $\rightarrow$ 0.667, sealing or export $\rightarrow$ 0.622 (Table 7, Fig. 4). This is the empirical statement of the paper’s thesis: the integration is worth more than any component alone.

Table 7: Integration ablation (Phase 6). Properties are means over 300 findings.

Config	Label	Severity	Faithful expl.	Sealed	Provenance	Export	Actionability
full	1	1	0.73	1	1	1	0.955
no_explanation	1	1	0.00	1	1	1	0.833
no_scoring	1	0	0.00	1	1	1	0.667
no_sealing	1	1	0.73	0	0	1	0.622
no_export	1	1	0.73	1	0	0	0.622

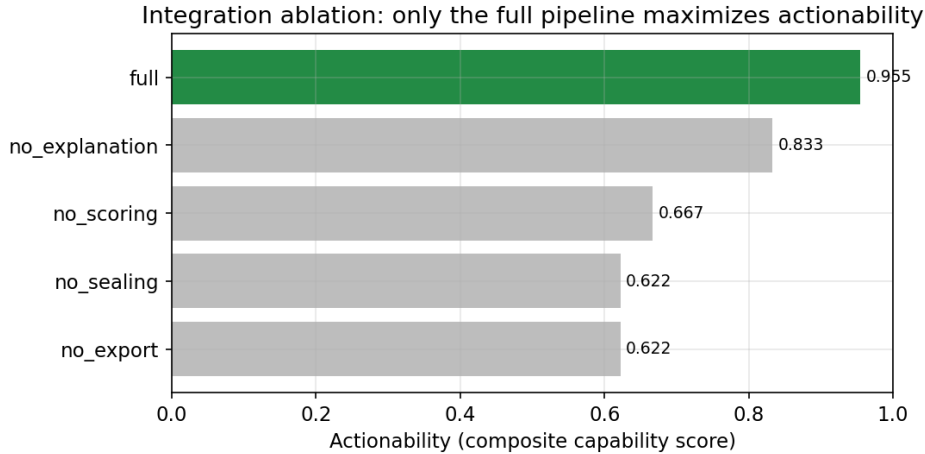


Figure 4: Actionability by configuration. Removing any module degrades the actionability of the emitted intelligence object; only the full pipeline is maximal.

1.7 Is the Evidence Sufficient? Discussion and Gaps

What is covered. All seven components produce real, positive results: strong text classification (0.919 macro-F1), working multilingual detection across six languages plus Hindi/Arabic cross-domain, a well-calibrated explainable scorer that nearly matches a black box while adding interpretability, 100% tamper detection at $\sim$ 8.7k items/s, valid STIX 2.1 export, and an ablation that empirically isolates the integration effect. This is sufficient to support the paper’s central *integration* claim and constitutes a complete, reproducible proof-of-concept.

What would strengthen it toward a top-tier SCI bar. (1) *Statistical validation*: add significance tests (McNemar for classifier pairs; Friedman+Nemenyi across models/languages)

and bootstrap 95% CIs; report mean $\pm$ std over multiple seeds, not single runs. (2) *Small-N languages*: DE ($N=24$), ES ($N=15$), and PT ($N=10$) pooled cells are too small for reliable claims (PT= 1.000 is an artifact of $N=10$); enlarge or report with CIs and explicit caveats. (3) *Stronger classification baselines*: include a transformer (XLM-R/BERT) head-to-head on CoDA text, not only classical TF-IDF, so the contribution is benchmarked against the modern state of the art. (4) *Explanation stability*: Jaccard 0.299 is low; investigate and report stability improvements or discuss the cause. (5) *Severity calibration figure*: add a reliability diagram and ECE alongside Brier. (6) *Clarify CIC-Darknet2020*: it is traffic, not text—frame it as a complementary modality so reviewers are not confused about the “threat-text” scope. With these additions the results move from a solid SCIE-level proof-of-concept to a defensible top-quartile submission.